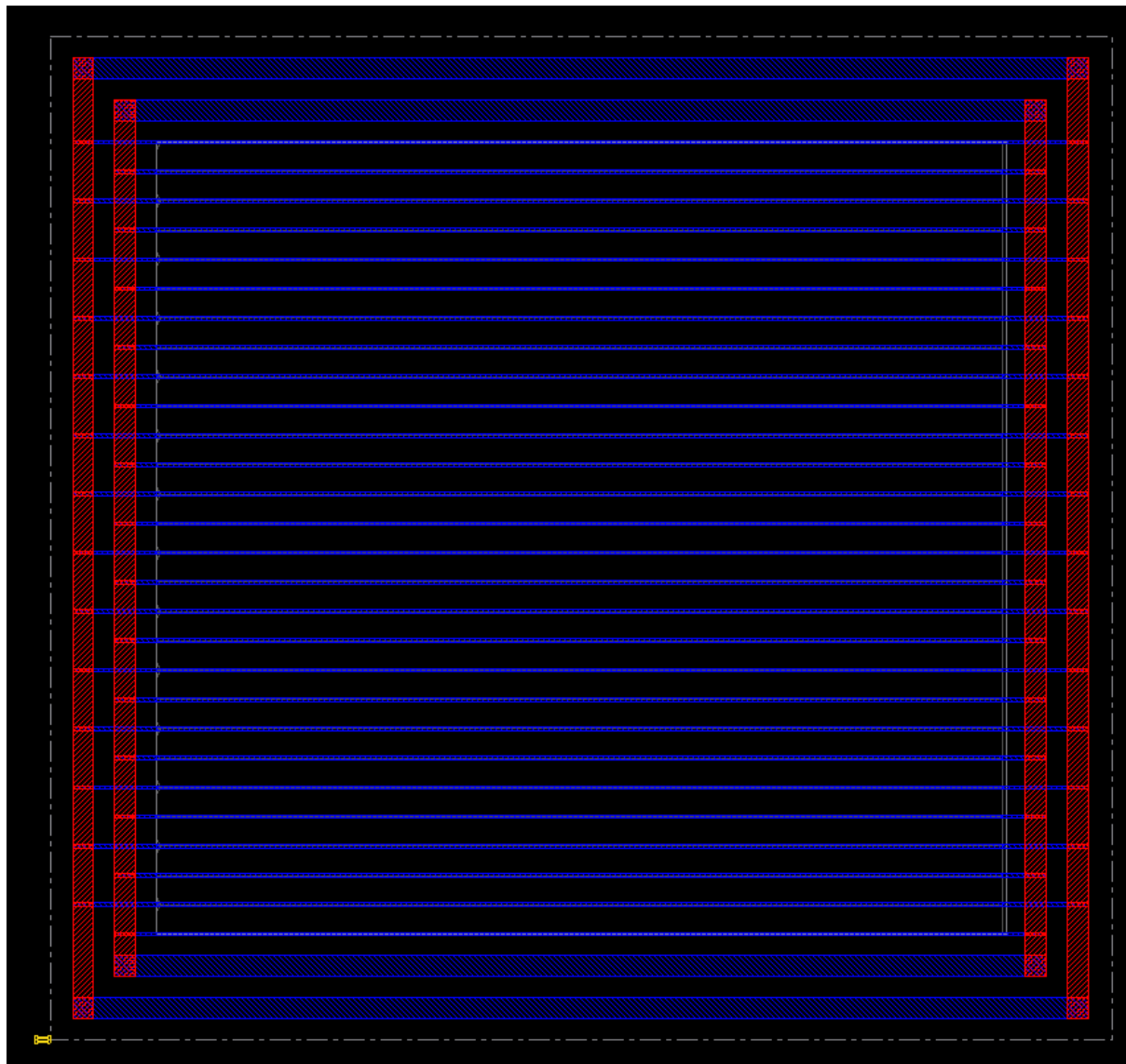


BJA64

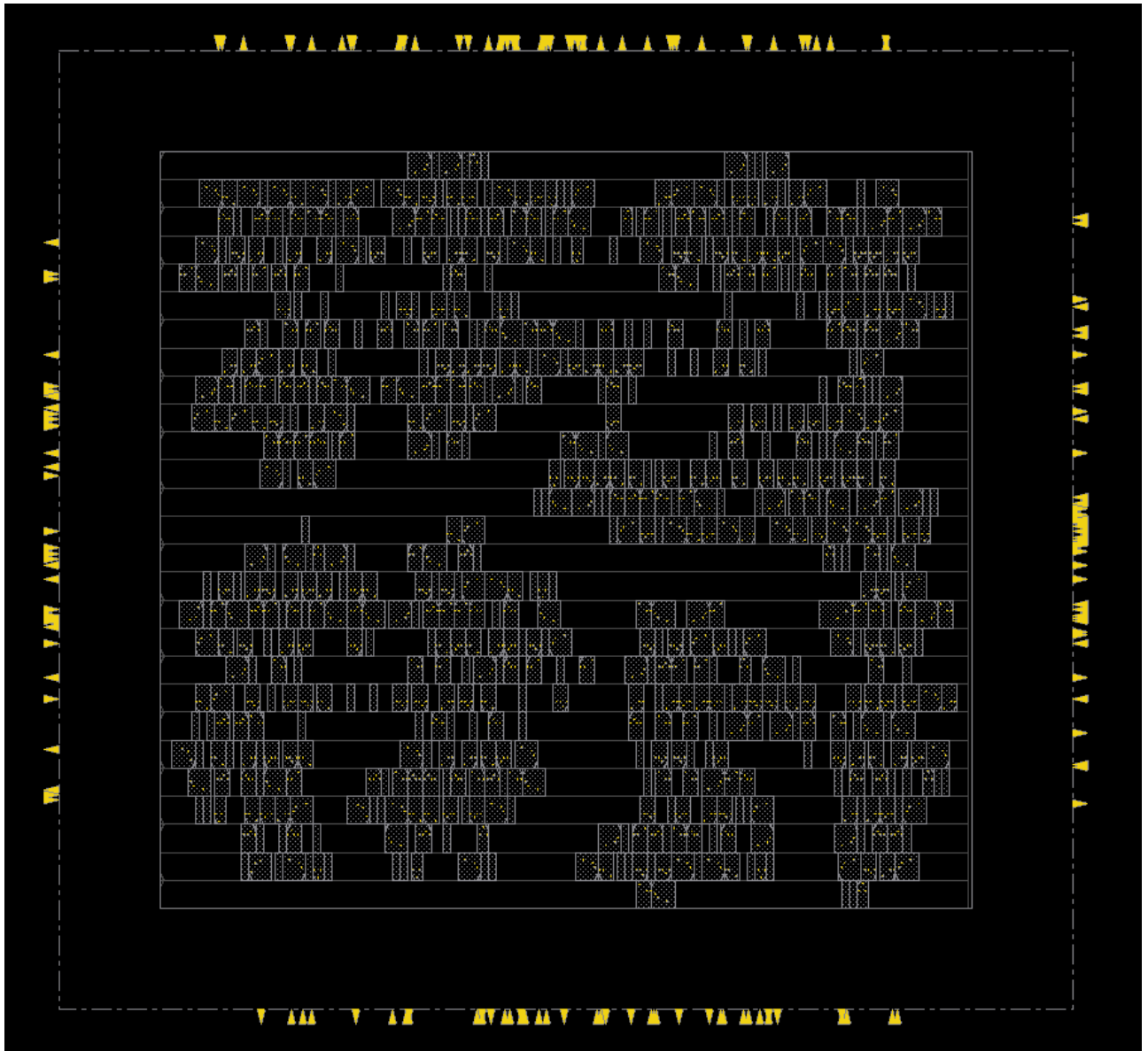
1. chip outlining



2. P/G network Design

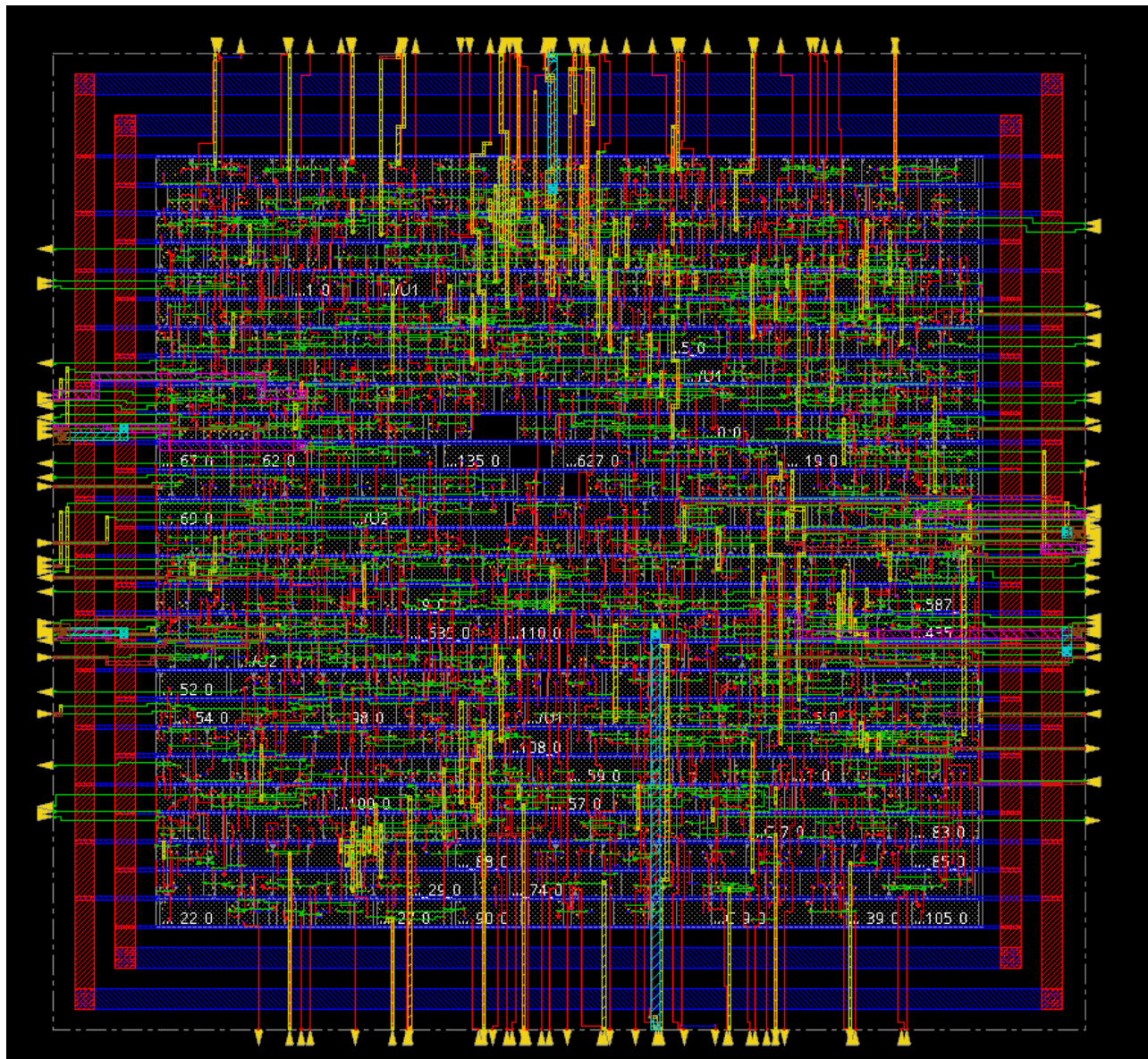


3. Placement



	WNS(ns)	TNS(ns)	Density(%)	Power(mW)	WL(um)
Placement	-0.360	-10.98	47.327	0.1851	4381
Pre-CTS opt	-0.067	-2.72	92.173	0.3526	
Post-CTS opt	-0.056	-1.883	93.94	0.3607	
Routing	-0.054	-1.768	93.94	0.3598	5499
Post-Route opt	-0.051	-1.778	96.12	0.3688	6190

4. Routing



5. Verification

a. verify DRC: 0 viols

```

Report VBR04.drc.rpt # string, default=, user setting
*** Starting Verify DRC (MEM: 1965.3) ***

VERIFY DRC ..... Starting Verification
VERIFY DRC ..... Initializing
VERIFY DRC ..... Deleting Existing Violations
VERIFY DRC ..... Creating Sub-Areas
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area: {0.000 0.000 50.540 47.880} 1 of 1
VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:00.1 ELAPSED TIME: 0.00 MEM: 256.1M) ***

```

i.

b. verify connectivity: 0 viols

i.

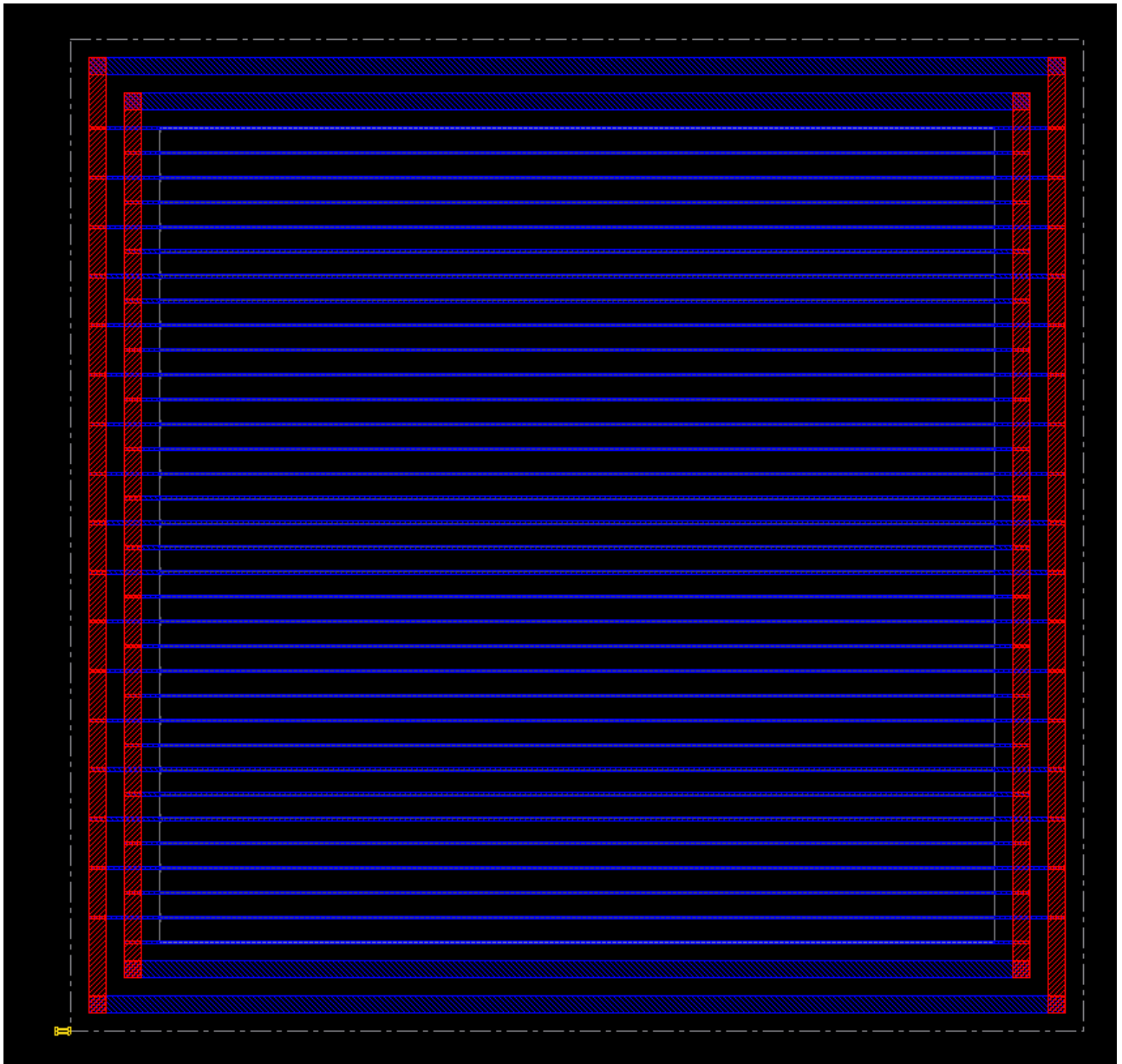
```
***** End: VERIFY CONNECTIVITY *****  
Verification Complete : 0 Viols. 0 Wrngs.  
(CPU Time: 0:00:00.0 MEM: 0.000M)
```

HCA64

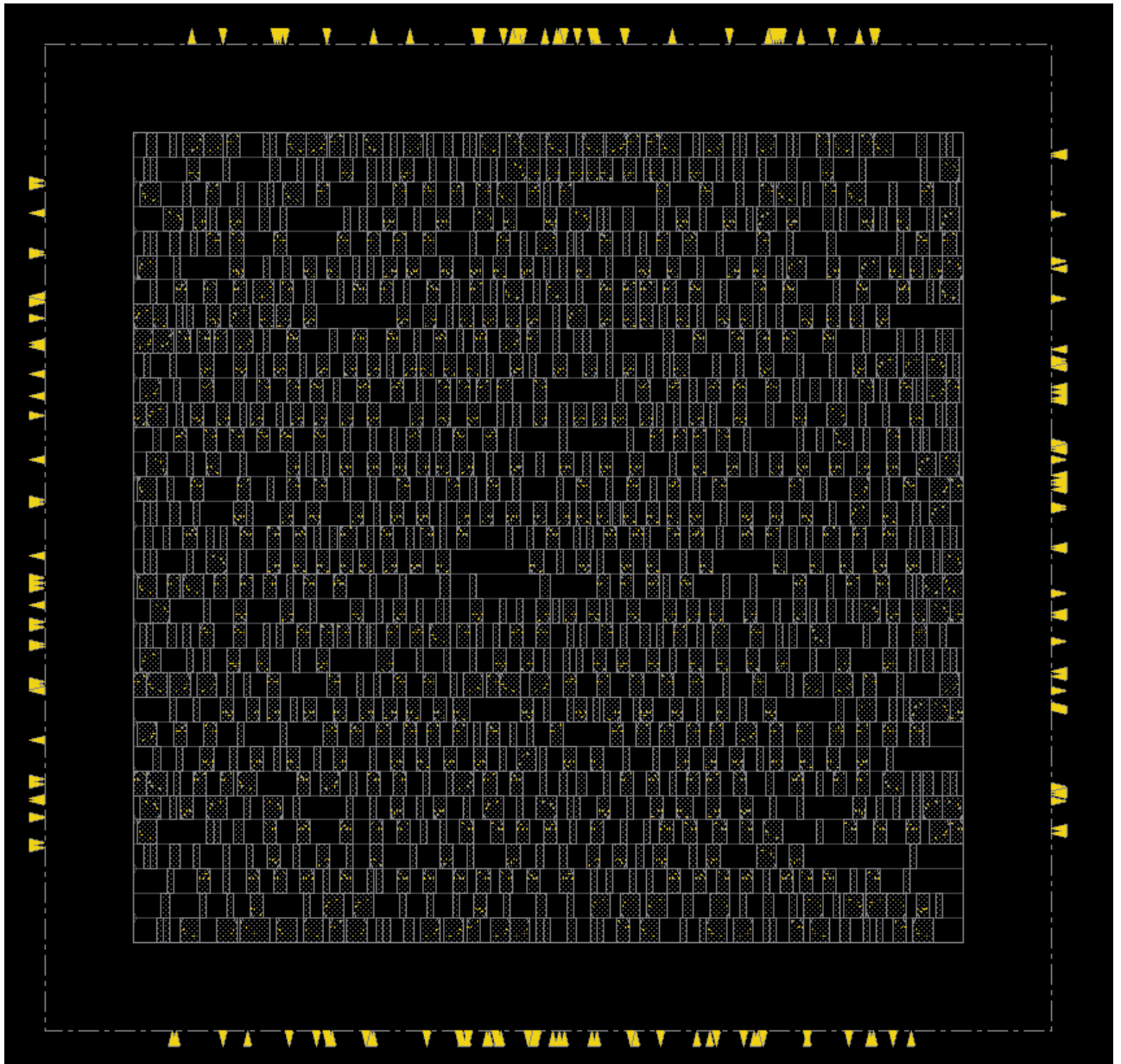
1. chip outlining



2. P/G network Design

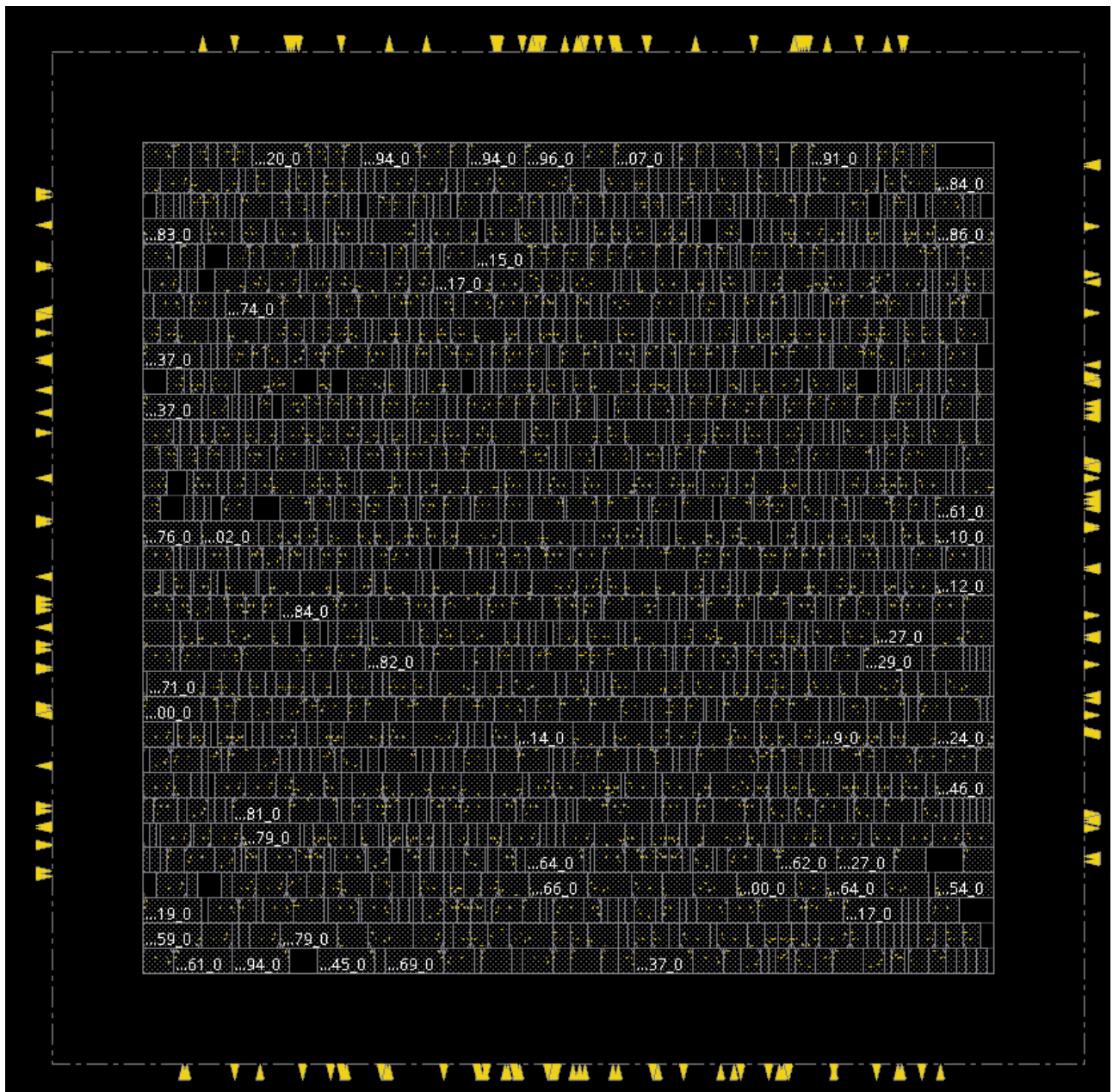


3. Placement



	WNS(ns)	TNS(ns)	Density(%)	Power(mW)	WL(um)
Placement	-0.176	-7.297	44.968	0.2491	7253
Pre-CTS opt	-0.026	-0.872	93.124	0.5219	
Post-CTS opt	-0.026	-0.804	93.952	0.5261	
Routing	-0.024	-0.677	93.952	0.5239	8185
Post-Route opt	-0.026	-0.692	95.57	0.5332	8829

4. Route



5. Verification

c. verify DRC: 1 viols

```
*** Starting Verify DRC (MEM: 1970.2) ***  
  
VERIFY DRC ..... Starting Verification  
VERIFY DRC ..... Initializing  
VERIFY DRC ..... Deleting Existing Violations  
VERIFY DRC ..... Creating Sub-Areas  
VERIFY DRC ..... Using new threading  
VERIFY DRC ..... Sub-Area: {0.000 0.000 57.400 56.280} 1 of 1  
VERIFY DRC ..... Sub-Area : 1 complete 1 Viols.  
  
Verification Complete : 1 Viols.  
  
Violation Summary By Layer and Type:  
  
                MetSpc  Totals  
metal2           1         1  
Totals           1         1
```

i.

d. verify connectivity: 0 viols

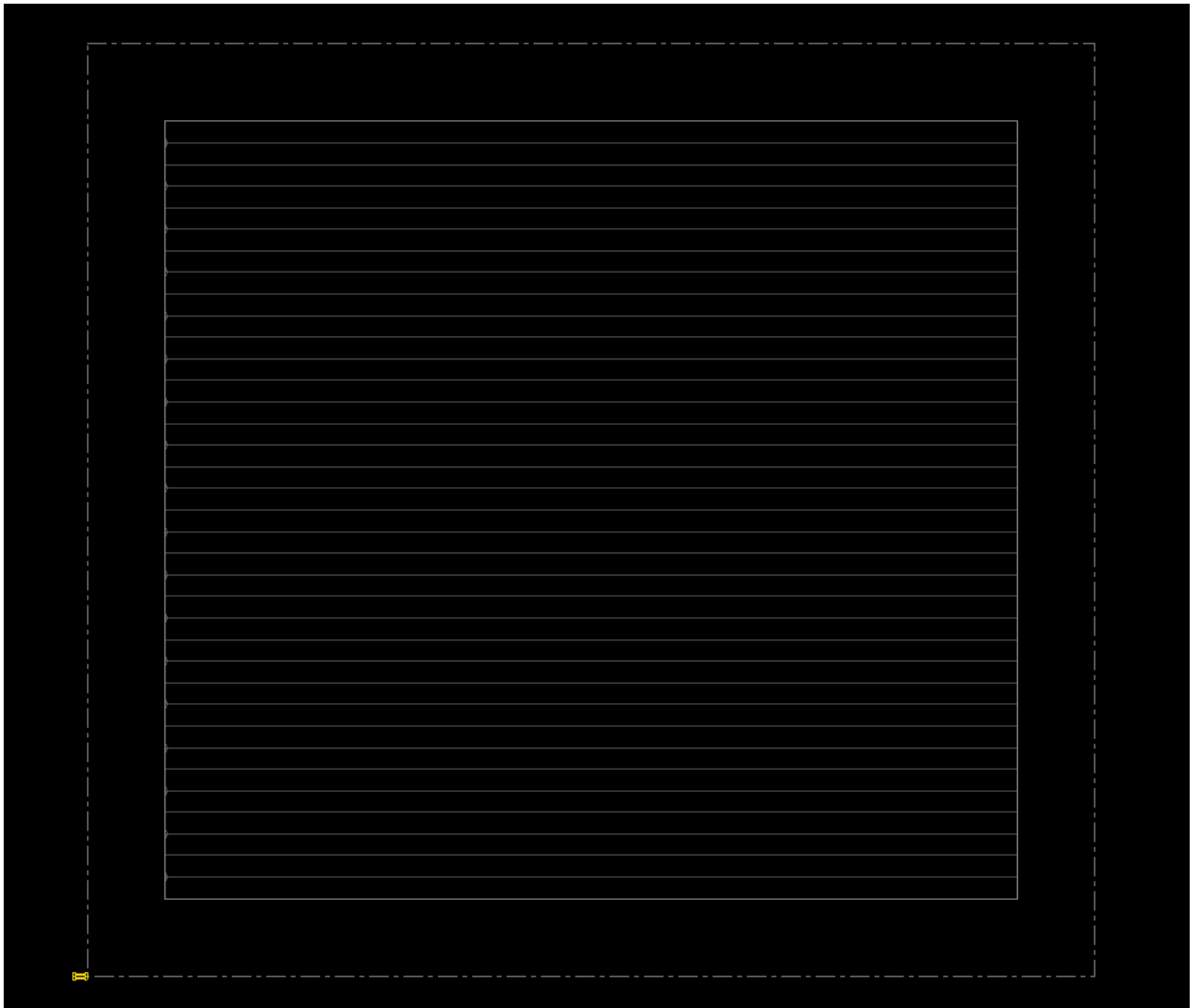
```
***** End: VERIFY CONNECTIVITY *****  
Verification Complete : 0 Viols. 0 Wrngs.  
(CPU Time: 0:00:00.0 MEM: 0.000M)
```

i.

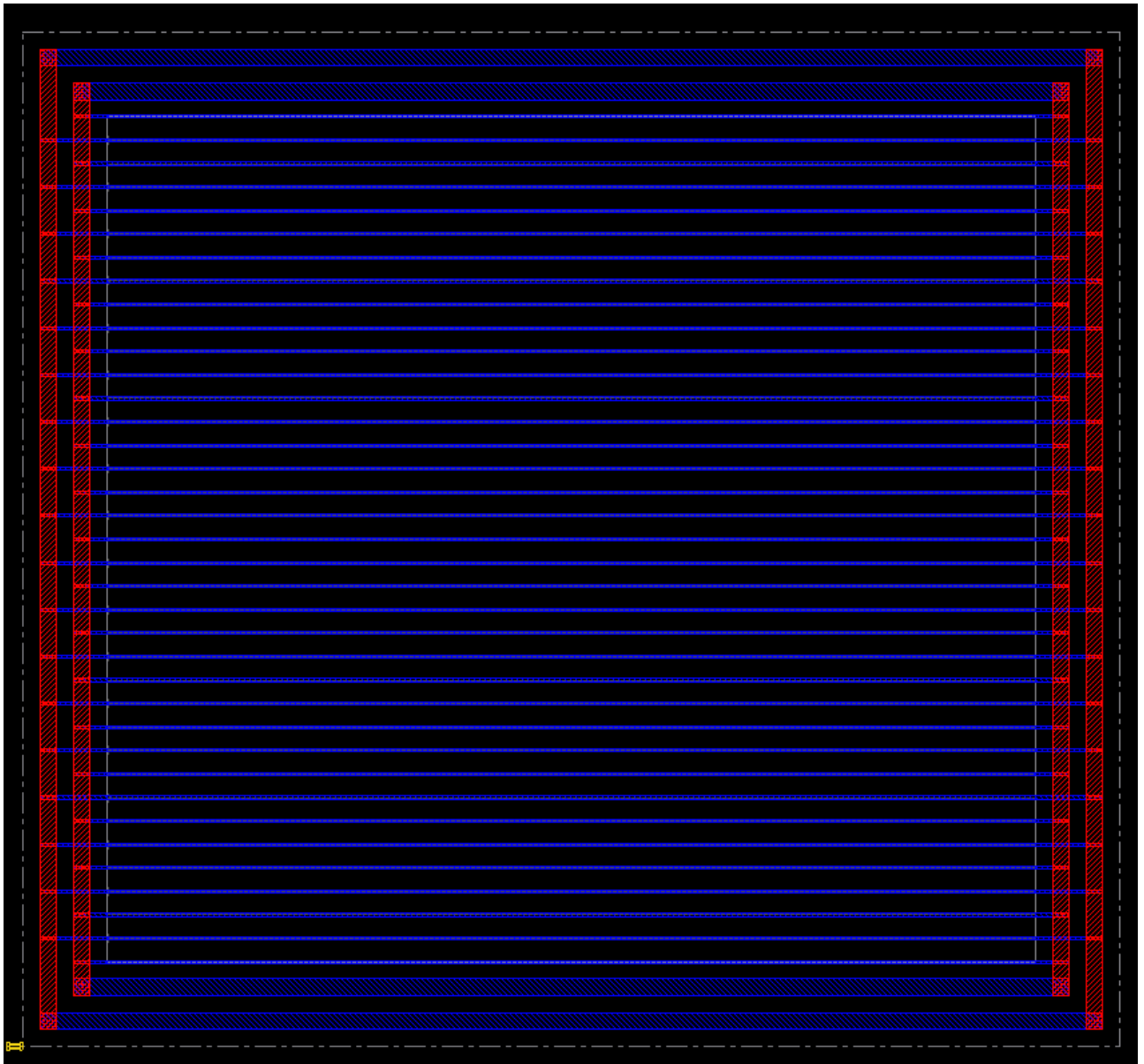
```
innovus 25> █
```

KSA64

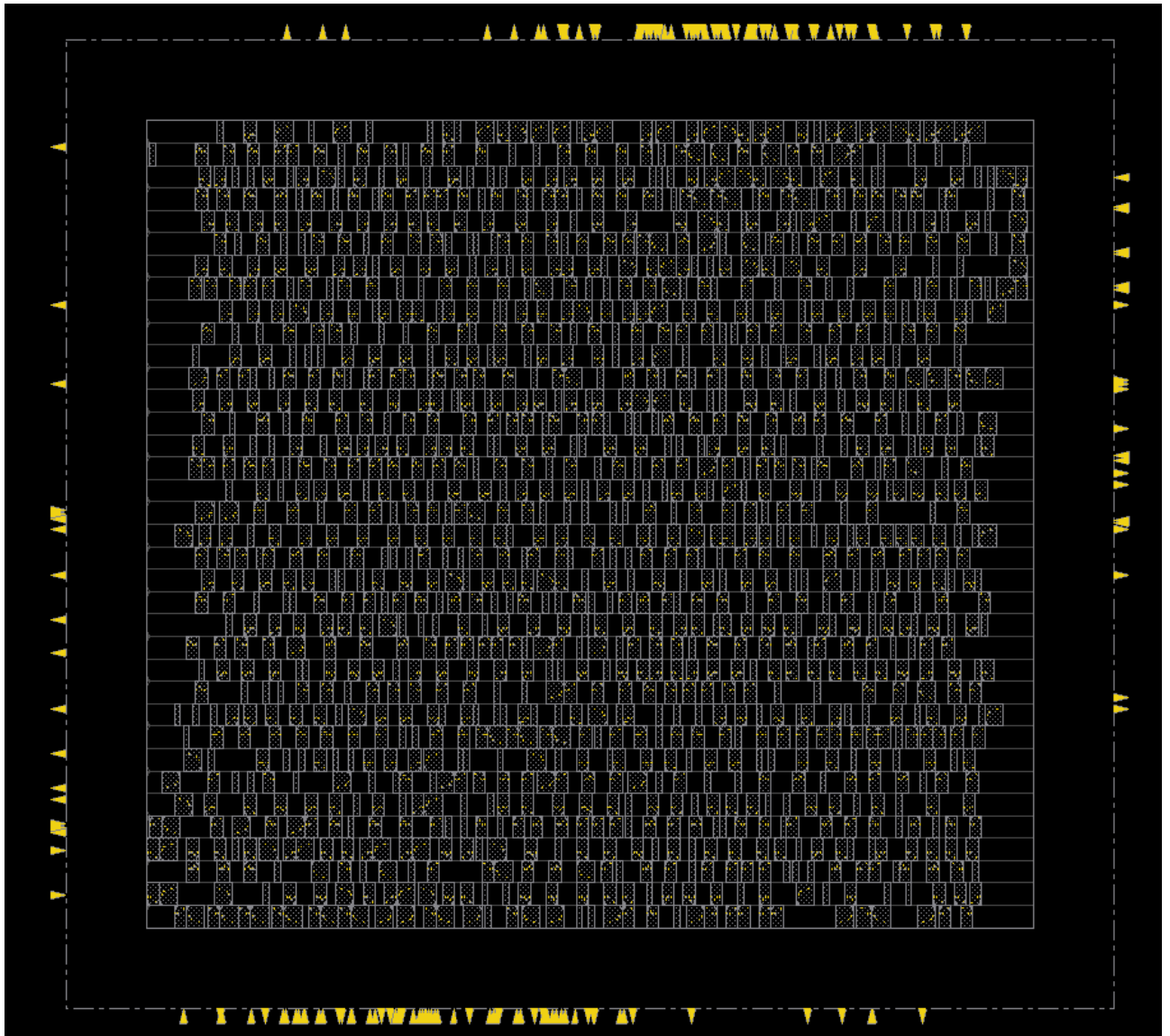
1. chip outlining



2. P/G network Design

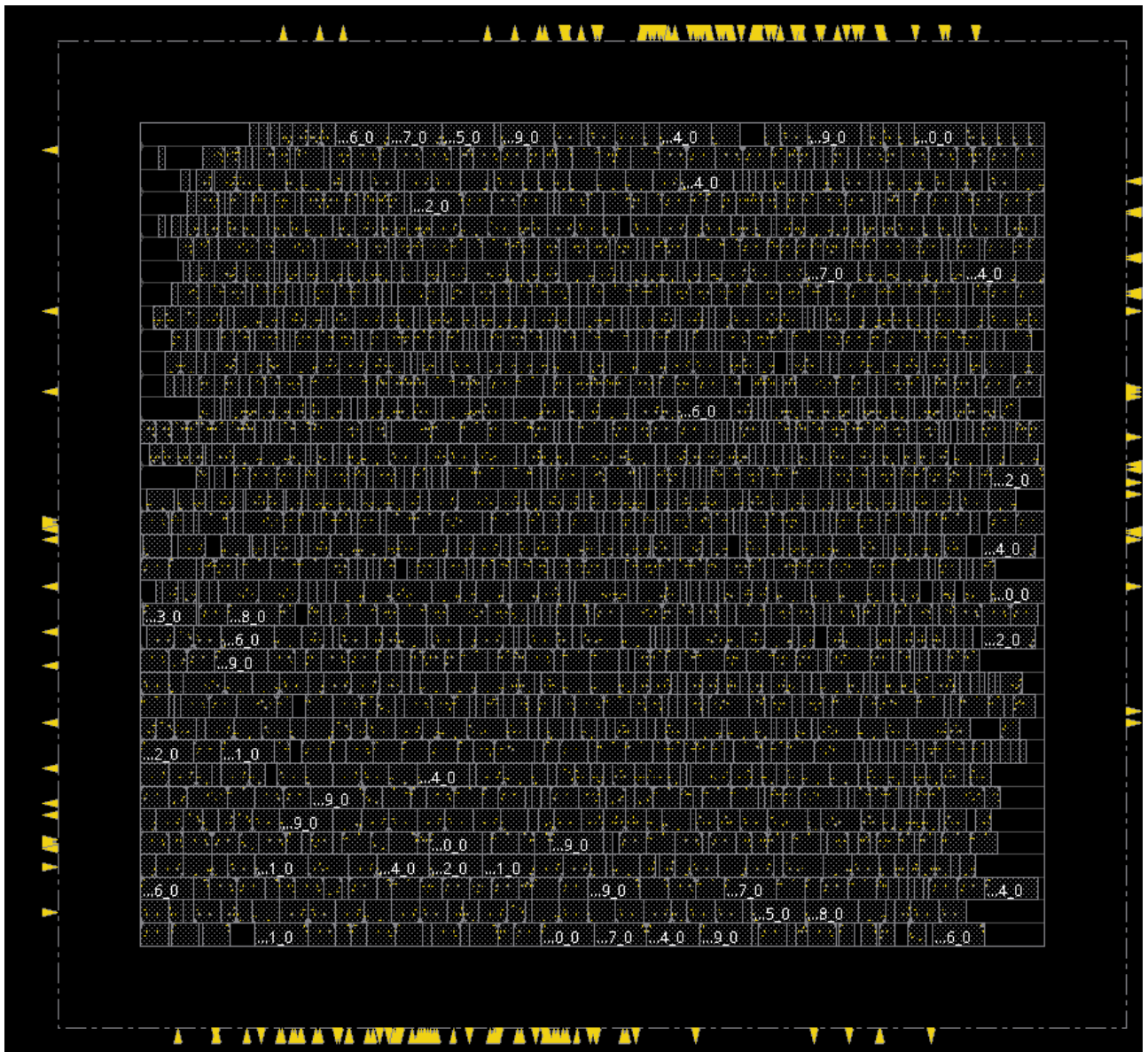


3. Placement



	WNS(ns)	TNS(ns)	Density(%)	Power(mW)	WL(um)
Placement	-0.166	-7.726	46.268	0.3377	11323
Pre-CTS opt	-0.014	-0.366	92.048	0.6860	
Post-CTS opt	-0.01	-0.215	94.311	0.7013	
Routing	-0.008	-0.096	94.311	0.6960	13254
Post-Route opt	-0.011	-0.131	95.246	0.7059	14635

4. Route



5. Verification

e. verify DRC: 4 viols

```

*** Starting Verify DRC (MEM: 2017.6) ***

VERIFY DRC ..... Starting Verification
VERIFY DRC ..... Initializing
VERIFY DRC ..... Deleting Existing Violations
VERIFY DRC ..... Creating Sub-Areas
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area: {0.000 0.000 33.480 60.480} 1 of 2
VERIFY DRC ..... Sub-Area : 1 complete 1 Viols.
VERIFY DRC ..... Sub-Area: {33.480 0.000 65.380 60.480} 2 of 2
VERIFY DRC ..... Sub-Area : 2 complete 3 Viols.

Verification Complete : 4 Viols.

Violation Summary By Layer and Type:

      Short  CutSpC  Totals
via1      0        1        1
metal2     1        0        1
metal8     2        0        2
Totals     3        1        4

*** End Verify DRC (CPU: 0:00:00.1 ELAPSED TIME: 0.00 MEM: 256.1M) ***

```

i.

f. verify connectivity: 0 viols

```
***** End: VERIFY CONNECTIVITY *****  
Verification Complete : 0 Viols. 0 Wrngs.  
(CPU Time: 0:00:00.0 MEM: 0.000M)
```

i.